

To Eun Kim

✉ toeunkim@cmu.edu

🎓 [Google Scholar](#)

🐙 <https://kimdanny.github.io/>

Education	Carnegie Mellon University (CMU) Aug 2023 - (2028) Ph.D. Language and Information Technology at Language Technologies Institute (LTI). 26/27: CMU LTI Presidential Fellowship (\$86,000) Advisor: Prof. Fernando Diaz Focus: Search Infrastructure for LLM Agents, Evaluation of Agents. University College London (UCL) Sep 2018 - Sep 2022 M.Eng. Computer Science (equivalent to a US Bachelor's & Master's degree) First-Class Honours Minor: Engineering Mathematics Thesis: Multi-Task Neural User Simulator for Task Oriented Dialogue System
Ongoing Research	Agentic Search Agent Artifact Reuse Simulated Benchmarking Complex Query Retrieval
Publications	Agent and Tool Search: Foundations, Techniques, and Open Challenges B. Wu, H.A. Rahmani, T.E. Kim , A. Mammadli, S. Qiao, X. Fu, V. Ramineni, X. Zhang, C. Meng, G. Drayson, A. Chowdhury, J. Ramos, A. Maiga, E. Yilmaz [paper] <i>Preprint</i> , 2026 Multi-Agent Transactive Memory T.E. Kim *, X. He*, D. Jain*, A. Agrawal, N. Arabzadeh, F. Diaz [paper] [code] <i>(COLM 2026) Lifelong Agents Workshop</i> AgentSearch: Indexing, Retrieval, and Ranking of AI Agents B. Wu, T.E. Kim , Y. Feng, F. Diaz, Z. Ren, E. Yilmaz [paper] [workshop page] <i>(SIGIR 2026) Workshop Organization</i> Evaluation of Information Access Agents under Simulated AI Marketplace Dynamics T.E. Kim , A. Salemi, H. Zamani, F. Diaz [paper] [code] <i>(SIGIR 2026) Perspectives</i> LTRR: Learning To Rank Retrievers for LLMs T.E. Kim , F. Diaz [paper] [code] <i>(SIGIR 2026) Short</i> <i>(SIGIR 2025) LiveRAG Workshop—Spotlight</i> Multilingual and Domain-Agnostic Tip-of-the-Tongue Query Generation for Simulated Evaluation X. He, T.E. Kim , F. Diaz, M. Fröbe, J. Arguello, B. Mitra [paper] [code] <i>(SIGIR 2026) Resource</i> Towards Reducible Uncertainty Modeling for Reliable Large Language Model Agents C. Oh, S. Park, T.E. Kim , J. Li, W. Li, S. Yeh, X. Du, H. Hassani, P. Bogdan, D. Song, S. Li [paper] <i>(ACL 2026) Main</i> MoR: Better Handling Diverse Queries with a Mixture of Sparse, Dense, and Human Retrievers J.S. Kalra*, X. Zhao*, T.E. Kim , F. Cai, F. Diaz, T. Wu [paper] [code] <i>(EMNLP 2025) Main</i> TeamCMU at Touché: Adversarial Co-Evolution for Advertisement Integration and Detection in Conversational Search T.E. Kim , J. Coelho, G. Onilude, J. Singh [paper] [code] <i>(CLEF 2025) Touché Lab—Best Paper Award</i>

Tip of the Tongue Query Elicitation for Simulated Evaluation
Y. He*, [T.E. Kim*](#), F. Diaz, J. Arguello, B. Mitra [\[paper\]](#) [\[code1\]](#) [\[code2\]](#)
(*SIGIR 2025*) *Resource*

Towards Fair RAG: On the Impact of Fair Ranking in Retrieval-Augmented Generation
[T.E. Kim](#), F. Diaz [\[paper\]](#) [\[code\]](#)
(*ICTIR 2025*) *Main*
(*NeurIPS 2024*) *AFME Workshop—Top-3 Spotlight*

Retrieval-Enhanced Machine Learning: Synthesis and Opportunities
[T.E. Kim](#), A. Salemi, A. Drozdov, F. Diaz, H. Zamani
(*SIGIR 2025 Tutorial*) (*SIGIR-AP 2024 Tutorial*) [\[paper\]](#)

Overview of the TREC 2025 Tip-of-the-Tongue Track
J. Arguello, F. Diaz, M. Fröbe, [T.E. Kim](#), B. Mitra [\[paper\]](#)
(*TREC 2025*) *Overview*

Overview of the TREC 2024 Tip-of-the-Tongue Track
J. Arguello, S. Bhargav, F. Diaz, [T.E. Kim](#), Y. He, E. Kanoulas, B. Mitra [\[paper\]](#)
(*TREC 2024*) *Overview*

A Comprehensive Survey on Generative Diffusion Models for Structured Data
H. Koo, [T.E. Kim](#) [\[paper\]](#)
Preprint, 2023

A Multi-Task Based Neural Model to Simulate Users in Goal-Oriented Dialogue Systems
[T.E. Kim](#), A. Lipani [\[paper\]](#) [\[poster\]](#) [\[code\]](#)
(*SIGIR 2022*) *Short*
(*SIGIR 2022*) *SCAI Workshop—Oral*

Condita: A State Machine Like Architecture for Multi-Modal Task Bots
J. Ramos*, [T.E. Kim*](#), Z. Shi, X. Fu, F. Ye, Y. Feng, A. Lipani [\[paper\]](#)
(*Alexa Prize TaskBot Challenge Proceedings 2022*) *System Report*

When and What to Ask Through World States and Text Instructions: IGLU NLP Challenge Solution
Z. Shi*, J. Ramos*, [T.E. Kim](#), X. Wang, H. Rahmani, A. Lipani [\[paper\]](#)
(*NeurIPS 2022*) *IGLU Competition—Research Prize Winner*

Attention-based Ingredient Phrase Parser
Z. Shi, P. Ni, M. Wang, [T.E. Kim](#), A. Lipani [\[paper\]](#) [\[code\]](#)
(*ESANN 2022*) *Main*

Research &
Industry
Roles

CMU Language Technologies Institute (LTI) Aug 2023 - (May 2028)
Graduate Research & Teaching Assistant Pittsburgh, USA

- Supervised by [Prof. Fernando Diaz](#).
- Courses taken: 11642 Search Engines (Prof. Jamie Callan); 10701 Intro to ML (Dr. Henry Chai); 11711 Advanced NLP (Prof. Graham Neubig); 11801 Quantitative Evaluation of Language Technologies (Prof. Fernando Diaz); 11797 Question Answering (Prof. Eric Nyberg, Prof. Teruko Mitamura); 11851 Talking to Robots (Prof. Yonatan Bisk); 11824 Subword Modeling (Prof. David Mortensen)
- TA: Quantitative Evaluation of Language Technologies

UMass Amherst Center for Intelligent Information Retrieval May 2026 - Aug 2026
Research Assistant Amherst, MA, USA

- Supervised by [Prof. Hamed Zamani](#).
- Training specialized retrievers for Deep Research agents.

UCL Web Intelligence Group May 2021 - Jun 2023

Research & Teaching Assistant UCL Centre for AI, London, UK

- Supervised by [Prof. Emine Yilmaz](#) and [Prof. Aldo Lipani](#)
- Leading the Alexa Prize TaskBot Challenge (May 2021 - Mar 2022). Engineered a Task-oriented Dialogue System by using Alexa Skills Kit, Evi, DynamoDB, Lambda, and CodePipeline. Constructed and maintained CI/CD pipelines, deployed multiple micro-services with Flask and Docker, dealing with thousands of user conversations with feedback loop.
- Five TA roles and four research papers.

Raft AI Jun 2022 - Jul 2022
Machine Learning Engineer Intern London, UK

- Tabular information extraction from OCR for freight forwarding automation.
- Use of modern dev tools, e.g., Gitlab, Jira, Sentry, Docker, Kubernetes, GCP.

UCL (Human-Computer) Interaction Centre - bachelor's group thesis Oct 2020 - Apr 2021
Project *Olfactory-NLP* with group members and [Prof. Marianna Obrist](#).

- Extracted smell-related phrases and clustered multiple passages into different olfactory groups.
- Literature Review: *An NLP Approach to Classify Smell Experiences*. [\[paper\]](#)
- *Exploring the potential of automating the process of clustering smell stories*. [\[paper\]](#) [\[code\]](#)

KB Securities Jun 2020 - Aug 2020
Data Analyst Intern Seoul, South Korea

- Built a mutual funds recommender system based on Network Analysis, Hierarchical Clustering and Association Rule Mining.
- Built a stock price prediction system using LSTM and news articles.
- Social media sentiment monitoring system using KB's NLP Engine and Automation software.

Royal College of General Practitioners (RCGP) Oct 2019 - Mar 2020
Software Engineer - UCL Industrial Exchange Network London, UK

- Implemented a tree-based task-oriented chatbot that assists in checking software compliance regimes of British healthcare organizations based on Dialogflow.

Awards

2026 CMU LTI ASA Presidential Fellowship – \$86,000
2025 CLEF Touché Lab Best Paper Award
2025 SIGIR NSF Student Travel Award – \$2000
2022 NeurIPS Interactive Grounded Language Understanding (IGLU) – Research Prize – \$650
2022 Automated Knowledge Graph Construction (AKBC) Conference – Registration fee waiver
2022 SIGIR Student Travel Grant – Registration fee waiver
2021 UCL Student Union best societies award – *Centenary Award (UCL AI Society)*
2021 UCL CS Undergraduate Summer Research Internship Fund – £3,200

Academic Service

Reviewer / Program Committee (PC):

- Conference: ARR August 2026, EVIA 2026, SIGIR 2026, ICLR 2026, ICLR 2023, CIKM 2022
- Journal: TOIS 2026 2025
- Workshop: ACL 2026 (CustomNLP4U), SIGIR 2026 (AgentSearch), EMNLP 2026, 2025 (Black-box NLP)

Tutorial Organization:

- SIGIR 2025 ([REML](#)), SIGIR-AP 2024 ([REML](#))

Workshop Organization:

- SIGIR 2026 ([AgentSearch](#))

Shared Task Organization:

- [NTCIR-19](#) Tip-of-the-Tongue (ToT) Track
- [TREC 2025](#), [TREC 2024](#) Tip-of-the-Tongue (ToT) Track

Invited Talks

AgentSearch Workshop at SIGIR 2026, *Agent Search and Marketplace Evaluation*, Jul 24th 2026. [\[slides\]](#)

NLP Interest Group at The Alan Turing Institute, *Can Dialogue Systems Have Theory-of-Mind Ability?*, Nov 24th 2022. [\[slides\]](#)

Knowledge Graphs Interest Group at The Alan Turing Institute (5th meet-up), *Personal Knowledge Graphs: Towards Better Conversational Assistants*, Oct 27th 2022. [\[slides\]](#)

Teaching Experience	CMU: (11-801) Quantitative Evaluation of Language Technologies, <i>TA</i> UCL: Intense Python Programming course, <i>Course Co-Organiser and Lecturer</i> UCL: (COMP0189) Applied Artificial Intelligence, <i>TA</i> UCL: (COMP0071) Software Engineering, <i>TA</i> UCL: (CEGE0004) Machine Learning for Data Science, <i>TA</i> UCL: (CEGE0096) Geospatial Programming, <i>TA</i> UCL Artificial Intelligence Society , <i>Lecturer</i>	Fall 2026 March 2023 Spring 2023 Spring 2023 Spring 2023 Fall 2022 2020 - 2022
Other Community Involvements	The Alan Turing Institute NLP & KG Interest Group, <i>member</i> The Alan Turing Institute Knowledge Graphs Interest Group, <i>member</i> Automated Knowledge Graph Construction (AKBC) Conference, <i>volunteer</i> UCL Artificial Intelligence Society , <i>Head of Tutorials Team, Lecturer</i> • Founder and lecturer of Machine Learning Tutorial Series • Season 20/21: [GitHub] [Lecture Recordings] Season 21/22: [GitHub] • Co-organised an international competition, <i>Climate Hack.AI</i> [GitHub] [media coverage] NLP paper reading group (Jiphyeonjeon) , <i>Season 2,3 Organizer</i> • Season 2: Survey on Task-Oriented Dialog System and Knowledge Graph construction. [slides] [recording] (spoken in Korean) • Season 3: Recent Advances and Challenges in Task-oriented Dialogue Systems. [slides] [recording] (spoken in Korean) Google Developer Student Club UCL Korean Society, <i>IT Director - webmail automation</i>	Nov 2022 Sep 2022 Nov 2022 Sep 2019 - Jun 2022 May 2021 - Nov 2022 Sep 2021 - Jun 2022 Jun 2020 - May 2021
Mentoring	Master's students • Dishank Jain, Xuhong He, Yiliu Yang (CMU MIIS) • Ambuj Agrawal (CMU MIIS; Now at Snowflake) • Yifan He (CMU MIIS; Now at Meta) CMU Paths to AI, <i>Mentor</i> CMU Graduate Application Support Program UCL CS Transition Mentor, (<i>helping 1st year CS students</i>)	Spring 2024, 2025 Fall 2023 Sep 2020 - Jun 2022
Professional Training	Google Developers Korea Machine Learning Bootcamp Deep Learning Specialization (Prof. Andrew Ng on Coursera) UCLQ Summer School in Quantum Technology Introduction to AI in the Data Center (NVIDIA) KAIST Artificial Intelligence Bootcamp	Oct 2020 - Jan 2021 Dec 2020 Sep 2020 Mar 2020 Jun 2019 - Aug 2019
Major Skills	Computer Language: Python, Java, C, Matlab, R, SQL, $\LaTeX$ Research Framework: PyTorch, TensorFlow, HuggingFace, Scikit-Learn SW Dev: Docker, Kafka, Node, Express, React, Flask Platform: Clustered Computing, Slurm, Unix, Git, AWS, GCP, Alexa Skills Kit Human Language: English (advanced), Korean (native), Chinese (intermediate)	
References	Available on request.	

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